

Friday 28/10/2019

Midterm 1

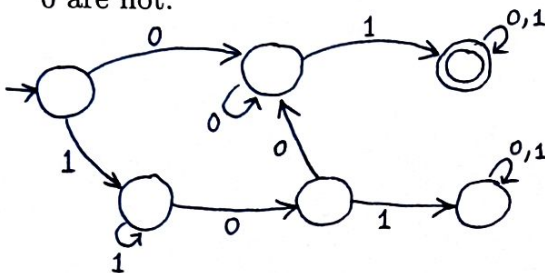
Duration: 90 minutes

Name: Solutions.

Student No:

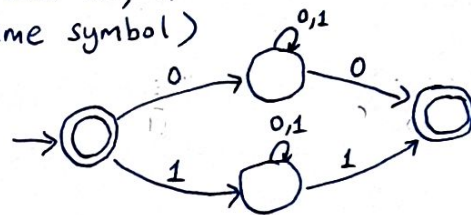
P1 [20 points] Construct finite automata (DFA or NFA) for the languages

- (a) $\{w \mid 01 \text{ is seen before } 101 \text{ in } w\}$ For example 001010 and 011 are accepted but 11010 and 0 are not.

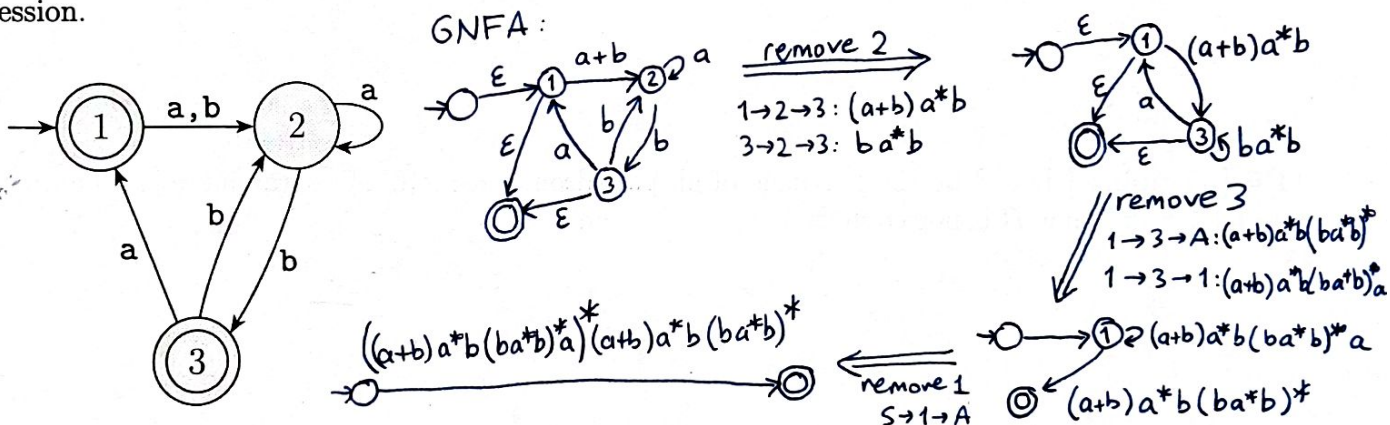


- (b) $\{w \mid w \text{ contains an equal number of occurrences of the substrings } 01 \text{ and } 10\}$.

(Observation: A string is in this language if and only if it starts and ends with the same symbol)



P2 [20 points] Use the procedure we saw in class to turn the following automata into a regular expression.



P3 [20 points] Let $\Sigma = \{1, \#\}$ and let $Y = \{w \mid w = x_1\#x_2\#\dots\#x_k \text{ for } k \geq 0, \text{ each } x_i \in 1^*, \text{ and } x_i \neq x_j \text{ for } i \neq j\}$. (For example, $111\#1\#11 \in Y$ but $11\#1\#11 \notin Y$.) Prove that Y is not regular.

Assume that Y is regular. Then, by Pumping Lemma, $\exists p$ such that $\forall s \in Y$, $|s| \geq p$, PL must hold. (i.e: s must be splittable to $s = xyz$ so that [1. $\forall i \ x y^i z \in Y$, 2. $|xy| \leq p$ 3. $|y| \geq 1$] holds.) Now, let $s = 1^p \# 1^{p+1} \# 1^{p+2} \# \dots \# 1^{p+p}$. Then, y must be within first p 1s. Say $y = 1^m$ for some $m \geq 1$. But then, when we pump up once, we get $s' = 1^{p+m} \# 1^{p+1} \# 1^{p+2} \# \dots \# 1^{2p}$, and since $p \geq m$, $p+m$ appears to be equal to one of $p+1, p+2, \dots, 2p$, and therefore $s' \notin Y$. Hence, our assumption was wrong, Y cannot be regular. $\square$

(Note: Choosing $s = 1^p \# 1^{p+p!}$ works, too.)

Then, if $y = 1^m$, pump up $\frac{p!}{m}$ times to get $s' \notin Y$.

P4 [20+20 bonus points] Give CFG and pushdown automata for the following languages:

- (a) $L = \{w | w \in \{0,1\}^*, w \text{ has odd length, and when even positions are deleted from } w, \text{ we obtain a palindrome.}\}$ For example, $1, 0011110 \in L$ but $\epsilon, 10010, 1010, 110 \notin L$.

CFG:

$$S \rightarrow OTSTO \mid 1TST1 \mid T$$

$$T \rightarrow 0 \mid 1$$

- (b) $\{a^i b^j c^k d^m \mid i + j = k + m\}$.

CFG:

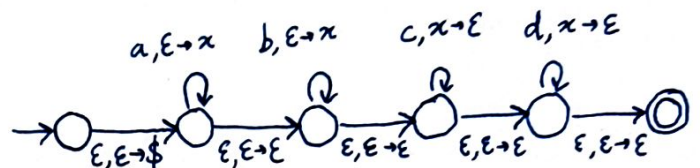
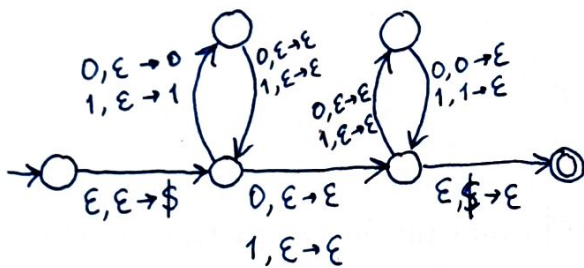
$$S \rightarrow ASB \mid \epsilon$$

$$A \rightarrow a \mid b$$

$$B \rightarrow c \mid d$$

PDA:

PDA:



P5 [20 points] Let B be the language of all palindromes over $\{0,1\}$ containing equal numbers of 0s and 1s. Show that B is not context free.

Assume B is context-free. Then, by Pumping Lemma, $\exists p$ such that

$\forall s (|s| \geq p) \in B$, s is splittable to $s = uvxyz$ so that 3 conditions hold:

1. $\forall i uv^i x y^i z \in B$
2. $|v| \geq 1$
3. $|vxy| \leq p$

Now, let $s = 0^p 1^{2p} 0^p$. Now, to satisfy all 3 conditions above,

- if we split with $\#0s$ and $\#1s$ are different in vy , we lose equality of 1s and 0s $s' \notin B$.

- if we split with $\#0s = \#1s$ in vy , it has to be like: $00 \dots 0 \overset{x}{1} \overset{y}{1} \dots 110000$
but then, when we pump up, palindrome property is lost. $s' \notin B$ again.

Therefore it is not possible to split s always. Hence, B is not context-free. $\square$